CHAPTER 16

Particle Accelerators

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This chapter describes the sources – accelerators – that can be used to obtain energetic charged particles, and secondarily neutrons and photons, that can induce nuclear reactions or chemical changes. Thus we include various accelerators for producing beams of charged particles as well as targets for neutron production. Electrons can be accelerated by the same principles as positive ions but in the following paragraphs we will focus on the acceleration of positive ions because these are more useful for nuclear reactions. Although accelerators can be used to produce neutrons, a more copious neutron source is the nuclear reactor, see Chapter 19. Small neutron sources, consisting of the spontaneous fissioning nuclide ^{252}Cf or of mixtures of elements like radium and beryllium in which nuclear transmutations (mostly (α,n)-reactions) produce neutrons, are discussed in §10.9.

16.1. CHARGED PARTICLE ACCELERATORS

In order to induce nuclear reactions with positively charged projectiles such as protons, deuterons, α-particles, oxygen ions, or uranium ions, it is necessary that the projectile particles have sufficient kinetic energy to overcome the Coulomb barrier created by the repulsion between the positive charges of the projectile and the target nucleus. While there is some probability that a positive projectile can tunnel through the Coulomb barrier at kinetic energies lower than the maximum value of the

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barrier, this probability is quite small until the kinetic energy is close to the barrier maximum.

As mentioned in §10.3, in 1919 Rutherford caused artificial transmutation of one element into another by using the α-particles emitted in the radioactive decay of ^{214}Po as projectiles. These α-particles had sufficient kinetic energy upon emission from the polonium nucleus (7.7 MeV) to overcome the Coulomb barrier and react with nitrogen nuclei. However, the kinetic energy of α-particles emitted in radioactive decay is generally insufficient to overcome the Coulomb barrier to react with nuclei of higher atomic numbers. Consequently, means had to be devised for acceleration of the charged projectiles to kinetic energies sufficient to achieve reaction.

The principle to be used to achieve the higher kinetic energies was obvious. The projectile particles would have to be ionized to obtain positively charged ions. If these ions could be accelerated through a potential difference of 1000 V, they would acquire 1000 eV additional kinetic energy, per unit of charge. If an α-particle of a +2 charge was to be accelerated through a potential difference of 10^6 V, it would acquire an additional kinetic energy or 2 MeV (eqn. (3.6)). The problem of obtaining the desired kinetic energy involved two aspects: first, the production of the charged particles: second, the acceleration through the necessary potential difference.

16.2. ION SOURCE

The problem of producing the positive ions is in principle relatively simple. If a gas is bombarded by energetic electrons, the atoms of the gas are ionized and positive ions produced. Figure 16.1 shows a simple ion source.

As hydrogen gas flows into the region above the filament, the electrons being emitted by the filament are accelerated to an anode (a typical voltage drop over the electrodes B_1—B_2 may be 100 V) and in their passage through the gas cause ionization. The positive ions are extracted by attraction to a negative electrode (the voltage drop over S_1—S_2 may be 1—10 kV) into the accelerator region. The vacuum at the beam extraction is of the order 10^{-4} Pa but in the ionization compartment it is normally 10^{-2} Pa. The basic principle, bombardment with electrons, is similar in the different types of ion sources that have been developed to meet the demands of the variety of ions and accelerator types in use. High ion currents of all elements in a variety of charge states can now be produced.

16.3. SINGLE-STAGE ACCELERATORS

The first successful accelerator for causing nuclear transmutations was developed by Cockcroft and Walton in 1932 and used a high voltage across an acceleration space.